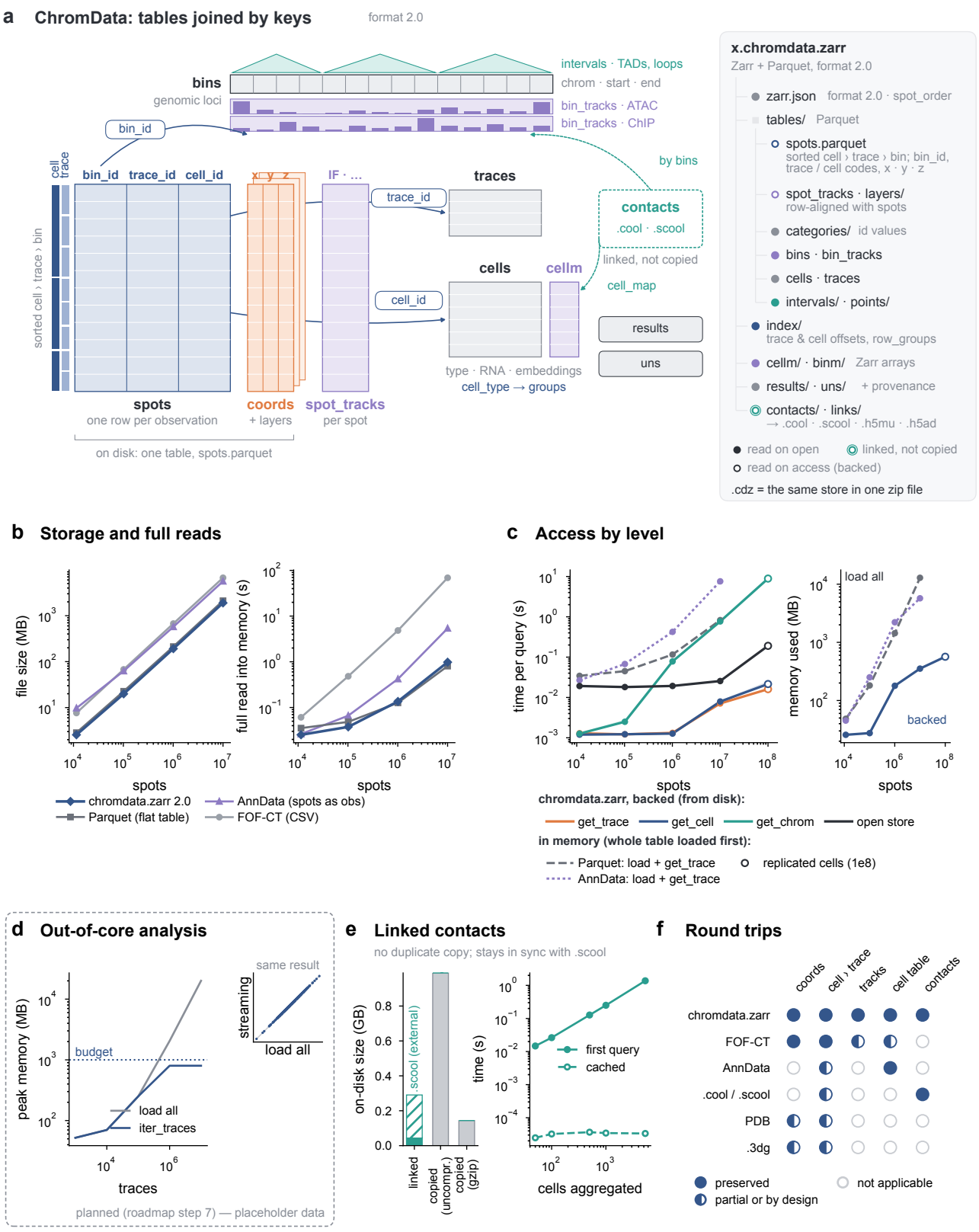

a
ChromData: tables joined by keys
format 2.0
intervals · TADs, loops
bins
chrom · start · end
genomic loci
bin_tracks · ATAC
bin_tracks · ChIP
bin_id
by bins
cell
trace
bin_id
trace_id
cell_id
x
y
z
IF · …
traces
trace_id
contacts
.cool · .scool
linked, not copied
cells
cellm
cell_map
cell_id
sorted cell › trace › bin
results
uns
type · RNA · embeddings
cell_type → groups
spots
one row per observation
coords
+ layers
spot_tracks
per spot
on disk: one table, spots.parquet
x.chromdata.zarr
Zarr + Parquet, format 2.0
zarr.json
format 2.0 · spot_order
tables/
Parquet
spots.parquet
sorted cell › trace › bin; bin_id, trace / cell codes, x · y · z
spot_tracks · layers/
row-aligned with spots
categories/
id values
bins · bin_tracks
cells · traces
intervals/ · points/
index/
trace & cell offsets, row_groups
cellm/ · binm/
Zarr arrays
results/ · uns/
+ provenance
contacts/ · links/
→ .cool · .scool · .h5mu · .h5ad
read on open
linked, not copied
read on access (backed)
.cdz = the same store in one zip file
b
Storage and full reads
file size (MB)
full read into memory (s)
spots
spots
chromdata.zarr 2.0
AnnData (spots as obs)
Parquet (flat table)
FOF-CT (CSV)
c
Access by level
time per query (s)
memory used (MB)
load all
backed
spots
spots
chromdata.zarr, backed (from disk):
get_trace
get_cell
get_chrom
open store
in memory (whole table loaded first):
Parquet: load + get_trace
replicated cells (1e8)
AnnData: load + get_trace
d
Out-of-core analysis
peak memory (MB)
budget
load all
iter_traces
traces
same result
streaming
load all
planned (roadmap step 7) — placeholder data
e
Linked contacts
no duplicate copy; stays in sync with .scool
on-disk size (GB)
.scool (external)
linked
copied (uncompr.)
copied (gzip)
time (s)
first query
cached
cells aggregated
f
Round trips
coords
cell › trace
tracks
cell table
contacts
chromdata.zarr
FOF-CT
AnnData
.cool / .scool
PDB
.3dg
preserved
partial or by design
not applicable